

Lecture 7: Forecasting Returns

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Outline

Forecasting Regressions

Dividend-Yield Forecasting



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Risk premia across assets

The **risk premium** of an asset, i , is defined as the **expected** excess return,

$$\mathbb{E}[\tilde{r}^i]$$

- ▶ **Linear Factor Models** (LFM's) describe how risk premia vary across assets.
- ▶ Most theories attribute the variation to difference in risks.



Example: CAPM

The CAPM says risk premia across different assets i are:

$$\mathbb{E}[\tilde{r}^i] = (\beta^{i,m}) \lambda_m$$

All risk premia are proportional (by beta) to the market risk premium.

- ▶ But the above form does not condition on time.
- ▶ The beta and both risk premia are estimated as stationary time series averages.



Risk premia over time

So how do risk premia change over time?

$$\mathbb{E}_t [\tilde{r}_{t+1}^i]$$

- ▶ Is the expected excess return of asset i always the same, no matter what period the investor considers?

$$\mathbb{E}_t [\tilde{r}_{t+1}^i] = \mathbb{E} [\tilde{r}^i]$$

- ▶ Or is the risk premium of an asset a function of time-varying factors, x_t ?

$$\mathbb{E}_t [\tilde{r}_{t+1}^i] = f(x_t)$$



Linear methods

If we believe risk premia vary over time, we must specify a functional form for $f(x)$ in

$$\mathbb{E}_t [\tilde{r}_{t+1}^i] = f(x_t)$$

as well as specifying the factor(s), x_t .

- ▶ If we specify a linear function, the statistics/numerics are much easier.
- ▶ Recall that a linear regression gives the best linear estimator of such a function $f(x)$.



Regressions to measure conditional expectations

Suppose

$$y = \alpha + \beta x + \epsilon$$

Then the expectation of y conditional on x is

$$\mathbb{E}[y | x] = \alpha + \beta x$$

Thus, if $\beta \neq 0$, the conditional expectation varies as x varies.



Forecasting regressions

A **forecasting** regression for returns is of the form:

$$\tilde{r}_{t+1}^i = \alpha + \beta x_t + \epsilon_{t+1}$$

- ▶ If $\beta \neq 0$, then the conditional expectation of $\tilde{r}_{t+1}^i$ varies over time as x_t varies.

$$\mathbb{E} [\tilde{r}_{t+1}^i | x_t] = \alpha + \beta x_t$$

- ▶ We similarly used regressions in LFM's to discover variation in risk premia across assets.



Classic view

**S&P has an 8% average excess return against the risk-free rate
no essence in trying to beat the market**

The **classic view** says risk premia are constant over time.

- ▶ Thus, in any forecasting regression of returns, $\beta = 0$.

$$\tilde{r}_{t+1}^i = \alpha + \beta x_t + \epsilon_{t+1}$$

- ▶ The classic view also says price growth is a **random walk** (with drift.)

$$\log P_{t+1} - \log P_t = \text{constant} + \epsilon_{t+1}$$

$$\text{So } \mathbb{E}_t \left[\frac{P_{t+1}}{P_t} \right] = \text{constant.}$$



Testing the classic view

**This is an AR model
use the past value of the variable
 r_t to predict r_{t+1}**

Test the classic view on the risk premium of the market index, λ_m .

- ▶ Consider using this period's return to forecast that of next period:

$$\tilde{r}_{t+1}^m = a + \beta \tilde{r}_t^m + \epsilon_{t+1}$$

- ▶ This test of the classic view of market return predictability uses the lagged return as the predictor variable.



Evidence: Is the market return autocorrelated?

$$r_{t+1} = a + \beta r_t + \epsilon_{t+1}$$

Table: Auto-regression estimates for market returns, excess market returns.

	Monthly		Annual	
	r^m	$\tilde{r}^m$	r^m	$\tilde{r}^m$
b	0.11	0.12	0.01	0.02
$t(b)$	2.02	2.05	0.09	0.15
R^2	0.01	0.01	0.00	0.00

- ▶ CRSP value-weighted equity markets, 1927-2010.
- ▶ CRSP 3-month U.S. treasury bill.
- ▶ GMM standard errors.



Evidence: Is the risk-free return autocorrelated?

$$r_{t,t+1}^f = a + \beta r_{t-1,t}^f + \epsilon_{t+1}$$

Table: Auto-regression estimates for the risk-free return.

	Monthly	Annual
b	0.89	0.92
$t(b)$	30.38	13.31
R^2	0.80	0.83

- ▶ CRSP value-weighted equity markets, 1927-2010.
- ▶ CRSP 3-month U.S. treasury bill.
- ▶ GMM standard errors.



Conclusions from the return auto-regressions

using lagged returns to make prediction is not good

The excess market return has a regression coefficient near zero, which fits the classic view of risk premia.

- ▶ High returns do not indicate particularly high or low returns going forward.
- ▶ The annual data estimates suggest stock returns, particularly excess returns, are i.i.d.
- ▶ The monthly data shows some autocorrelation, but not much explanatory power.
- ▶ Furthermore, trading costs would seem to make this small predictability a novelty of no economic importance.



Other Ways to See Predictability?

For many years, academics and practitioners have found these same results.

- ▶ This reinforced classic view that returns are unpredictable—that prices are essentially a random walk.
- ▶ But even if past returns do not predict future returns, how about some other predictor, x_t ?
- ▶ How about forecasting at longer horizons?



Outline

Forecasting Regressions

Dividend-Yield Forecasting



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Signals

Notwithstanding the “classic” view, asset managers use many signals to try to forecast returns with linear regression.

- ▶ Macroeconomic signals
- ▶ Asset return signals
- ▶ Short-term signals / forecast horizons
- ▶ Long-term signals / forecast horizons

The dividend-price ratio, (also known as the dividend-yield,) is one of the most famous examples.



Dividend-yield

The **dividend-yield** DP_t refers to the **dividend-price** ratio, $\frac{D_t}{P_t}$.

- ▶ Other common cash-flow-to-value measures include earnings-price and book-price (book-market) ratios.
- ▶ Obviously, using value-to-cashflow ratios such as dividend-price works the same.
- ▶ For an individual stock, dividends are not paid continuously, but for the market index, there is a steady stream for analysis.



Returns and the dividend yield

By definition, stock returns are

$$R_{t+1} \equiv \frac{P_{t+1} + D_{t+1}}{P_t}$$
$$R_{t+1} \equiv \left(\frac{D_t}{P_t} \right) \frac{D_{t+1}}{D_t} + \frac{P_{t+1}}{P_t}$$

This identity holds for horizon, $t + k$, and in expectation:

$$\mathbb{E}_t [R_{t,t+k}] = \text{DP}_t \mathbb{E}_t \left[\frac{D_{t+k}}{D_t} \right] + \mathbb{E}_t \left[\frac{P_{t+k}}{P_t} \right]$$



Classic view of dividend yield

In the classic view of risk premia,

- ▶ Expected returns are constant: $\mathbb{E}_t \left[r_{t,t+k} \right] = \theta_r$.
- ▶ Price appreciation is a random walk, $\mathbb{E}_t \left[\frac{P_{t+k}}{P_t} \right] = \theta_p$.

$$\theta_r = \text{DP}_t \mathbb{E}_t \left[\frac{D_{t+k}}{D_t} \right] + \theta_p$$

So under the classic view, **in brackets is dividend growth**

- ▶ An increase in the dividend-yield is offset by a decrease in expected dividend growth.



Evidence: Stock-market Predictability

The R^2 for 5-years is too high
there should be more factors that contribute to the fact that the return increases
factors other than what's happening to the DP

$$\tilde{r}_{t,t+k}^m = a + \beta DP_t + \epsilon_{t+k}$$

Table: Stock Return Predictability Regressions.

	Horizon		
	1 month	1 year	5 years
b	0.25	4.08	21.27
$t(b)$	1.01	2.45	4.43
R^2	.01	.09	.31

Regression of cumulative excess returns on dividend-price ratio.

- ▶ NYSE/AMEX/NASDAQ value-weighted equity markets.
- ▶ Monthly data, 1927-2010.
- ▶ GMM standard errors.



Interpreting the regression estimates

At a one-month horizon,

- ▶ Slope coefficient is insignificant—statistically and economically.
- ▶ Agrees with the implications of the auto-regression.
- ▶ More evidence seemingly supportive of the classic view.

At longer horizons,

- ▶ Coefficient is economically significant.
- ▶ At one-year, a one-point increase in dividend-price forecasts a four-point increase in returns!



Illustration of return predictability

high correlation between DP and return

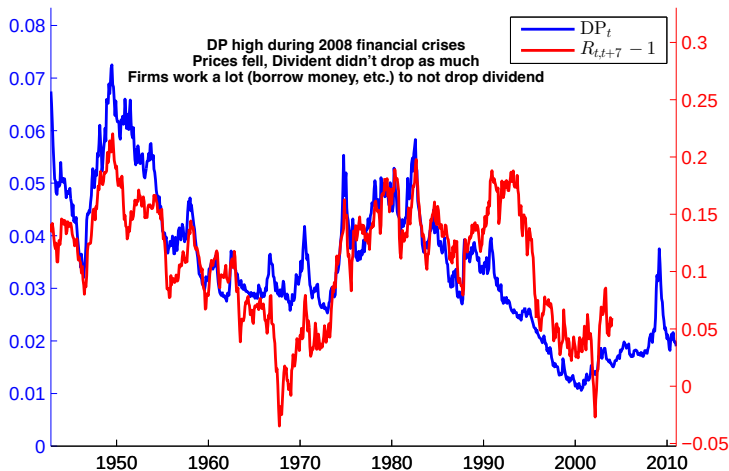


Figure: Market dividend-price ratio plotted against the next seven years cumulative return. CRSP monthly data, 1927-2010.



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Modern view of dividend yield

DP and returns are increasing together

The empirical evidence above shows:

- ▶ Expected returns increase one-for-one with the dividend-yield.
- ▶ This is not offset by dividend growth or price appreciation.
- ▶ Instead, estimates show prices move the wrong way—**increase** expected returns even more.

$$\mathbb{E}_t [R_{t,t+k}] = DP_t \mathbb{E}_t \left[\frac{D_{t+k}}{D_t} \right] + \mathbb{E}_t \left[\frac{P_{t+k}}{P_t} \right]$$



When prices are low, we (used to / now) expect...

Classic view: if prices are down (blue dashes), dividend will go lower to price levels (red dashes)

Modern view: if there's a gap between dividend and price levels, dividend will maintain its levels / stay the same (purple dashes) and prices will rebound to meet the price levels (green dashes)

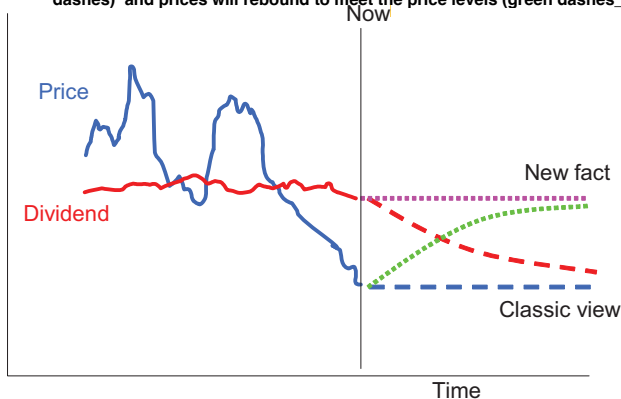


Figure: Source: Cochrane (2012)



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Long Horizons as a Way to See Predictability

Predictability of market returns by dividend-yield only seen in long horizons regressions.

- ▶ Due to persistent nature of the forecasting variable, DP_t .
- ▶ Autoregressive coefficient at a monthly frequency is about .98!

$$DP_{t+1} = a + b DP_t + \epsilon_{t+1}$$



Other Forecasting Variables?

Other variables seem to have similar ability to forecast returns.

- ▶ Cyclically-adjusted price-earnings ratios
- ▶ Macro-economic indicators. (Investment, consumption, etc.)
- ▶ Inflation and rates. (The “Fed Model”)



Statistical concerns

The dividend-price predictability is controversial.

- ▶ DP is a persistent variable, (high autocorrelation.)
- ▶ Regressions where x has high autocorrelation can be biased or mis-specified.

This is a very active area of research to find the best predicting variables and models.



References

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